Exercise 3

In this exercise, you will analyse a dataset obtained from the London transport system (TfL). The data is in a filled called `tfl_readership.csv` (comma-separated-values format). As in Exercise 2, we will load and view the data using `pandas`.

```
# If you are running this on Google Colab, uncomment and run the
following lines; otherwise ignore this cell
# from google.colab import drive
# drive.mount('/content/drive')

import math
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Load data
df_tfl = pd.read_csv('tfl_ridership.csv')
# If running on Google Colab change path to
'/content/drive/MyDrive/IB-Data-Science/Exercises/tfl_ridership.csv'

df_tfl.head(13)
```

```
       Year Period       Start         End Days  Bus cash (000s)  \
0   2000/01   P 01  01 Apr '00  29 Apr '00  29d              884
1   2000/01   P 02  30 Apr '00  27 May '00  28d              949
2   2000/01   P 03  28 May '00  24 Jun '00  28d              945
3   2000/01   P 04  25 Jun '00  22 Jul '00  28d              981
4   2000/01   P 05  23 Jul '00  19 Aug '00  28d              958
5   2000/01   P 06  20 Aug '00  16 Sep '00  28d              984
6   2000/01   P 07  17 Sep '00  14 Oct '00  28d             1001
7   2000/01   P 08  15 Oct '00  11 Nov '00  28d              979
8   2000/01   P 09  12 Nov '00  09 Dec '00  28d              971
9   2000/01   P 10  10 Dec '00  06 Jan '01  28d              912
10  2000/01   P 11  07 Jan '01  03 Feb '01  28d              943
11  2000/01   P 12  04 Feb '01  03 Mar '01  28d              975
12  2000/01   P 13  04 Mar '01  31 Mar '01  28d              974

    Bus Oyster PAYG (000s)  Bus Contactless (000s)  \
0                        0                        0
1                        0                        0
2                        0                        0
3                        0                        0
4                        0                        0
5                        0                        0
6                        0                        0
7                        0                        0
8                        0                        0
```

9	0	0
10	0	0
11	0	0
12	0	0

	Bus One Day Bus Pass (000s)	Bus Day Travelcard (000s)	... \
0	210	231	...
1	214	205	...
2	209	221	...
3	216	241	...
4	225	248	...
5	243	236	...
6	205	216	...
7	199	221	...
8	184	212	...
9	192	211	...
10	193	186	...
11	194	210	...
12	186	204	...

	Tube Contactless (000s)	Tube Day Travelcard (000s) \
0	0	655
1	0	605
2	0	650
3	0	708
4	0	730
5	0	702
6	0	639
7	0	668
8	0	640
9	0	631
10	0	556
11	0	617
12	0	584

	Tube Season Travelcard (000s)	Tube Other incl free (000s) \
0	1066	200
1	1168	217
2	1154	212
3	1196	214
4	1165	165
5	1164	151
6	1286	196
7	1298	220
8	1302	242
9	993	195
10	1259	234
11	1237	246
12	1262	266

```
    Tube Total (000s)  TfL Rail (000s)  Overground (000s)  DLR (000s)
\
0                2509               0                  0          96

1                2598               0                  0          93

2                2623               0                  0          98

3                2761               0                  0         105

4                2643               0                  0         103

5                2608               0                  0         100

6                2763               0                  0         107

7                2819               0                  0         113

8                2839               0                  0         114

9                2359               0                  0          90

10               2634               0                  0         110

11               2688               0                  0         120

12               2699               0                  0         119


    Tram (000s)  Air Line (000s)
0          45.8              0.0
1          46.5              0.0
2          47.1              0.0
3          50.8              0.0
4          50.3              0.0
5          49.2              0.0
6          48.8              0.0
7          51.5              0.0
8          54.0              0.0
9          55.3              0.0
10         50.1              0.0
11         50.5              0.0
12         47.7              0.0

[13 rows x 26 columns]
```

Each row of our data frame represents the average daily ridership over a 28/29 day period for various types of transport and tickets (bus, tube etc.). We have used the `.head()` command to display the top 13 rows of the data frame (corresponding to one year). Focusing on the "Tube Total" column, notice the dip in ridership in row 9 (presumably due to Christmas/New Year's), and also the slight dip during the summer (rows 4,5).

```
#df_tfl.sample(3)  #random sample of 3 rows
df_tfl.tail(3)  #last 3 rows
```

```
        Year Period       Start         End Days  Bus cash (000s)  \
242  2018/19   P 09  11 Nov '18  08 Dec '18  28d                0
243  2018/19   P 10  09 Dec '18  05 Jan '19  28d                0
244  2018/19   P 11  06 Jan '19  02 Feb '19  28d                0

     Bus Oyster PAYG (000s)  Bus Contactless (000s)  \
242                    1110                     1089
243                    1001                      949
244                    1036                     1075

     Bus One Day Bus Pass (000s)  Bus Day Travelcard (000s)  ...  \
242                            0                         41  ...
243                            0                         38  ...
244                            0                         30  ...

     Tube Contactless (000s)  Tube Day Travelcard (000s)  \
242                     1399                        249
243                     1110                        242
244                     1310                        204

     Tube Season Travelcard (000s)  Tube Other incl free (000s)  \
242                           1017                          334
243                            632                          259
244                            924                          305

     Tube Total (000s)  TfL Rail (000s)  Overground (000s)  DLR (000s)
\
242               4221              996                557         355

243               3279              750                414         270

244               3809              929                517         333


     Tram (000s)  Air Line (000s)
242         84.1              2.6
243         66.3              3.2
244         79.3              2.3

[3 rows x 26 columns]
```

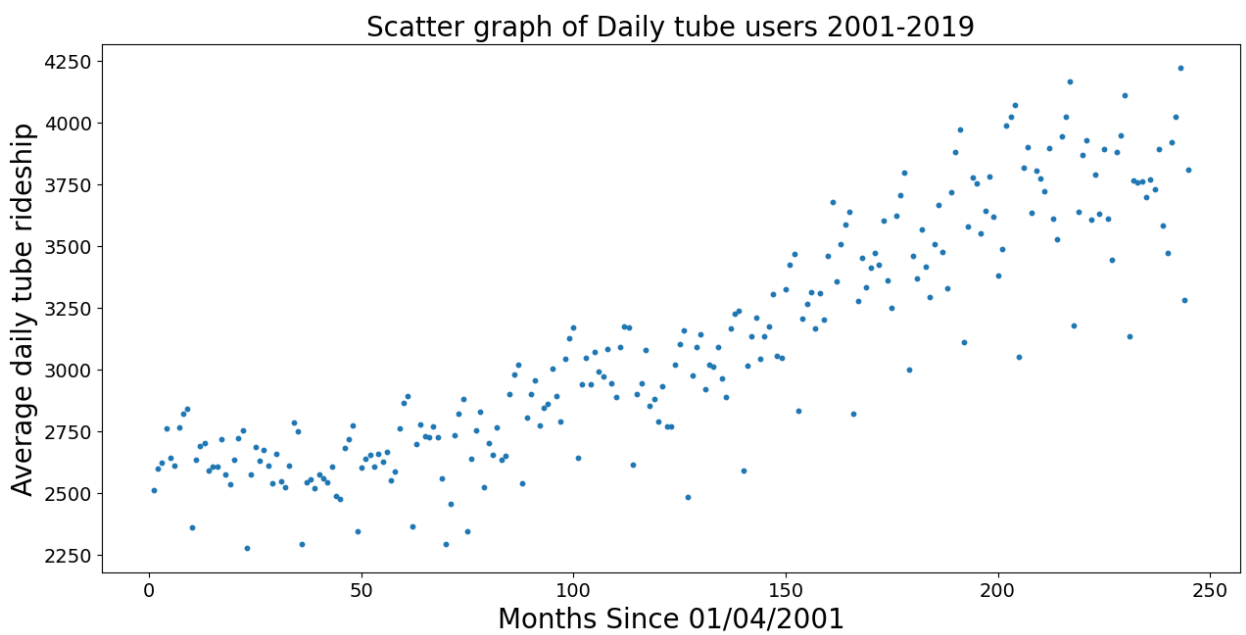
The dataframe contains $N = 245$ counting periods (of 28/29 days each) from 1 April 2000 to 2 Feb 2019. We now define a numpy array consisting of the values in the ' Tube Total (000s)' column:

```
yvals = np.array(df_tfl['Tube Total (000s)'])
N = np.size(yvals)
xvals = np.linspace(1,N,N) #an array containing the values 1,2....,N
```

We now have a time series consisting of points (x_i, y_i), for $i=1,\ldots,N$, where y_i is the average daily tube rideship in counting period $x_i=i$.

3a) Plot the data in a scatterplot

```
import matplotlib.pyplot as plt

plt.rcParams['figure.figsize'] = [15, 7]
plt.rcParams['axes.titlesize'] = 20
plt.rcParams['axes.labelsize'] = 20
plt.rcParams['xtick.labelsize'] = 14
plt.rcParams['ytick.labelsize'] = 14

plt.scatter(xvals, yvals, s=10)   # s can be used to adjust the size
of the dots
plt.xlabel('Months Since 01/04/2001')
plt.ylabel('Average daily tube rideship')
plt.title('Scatter graph of Daily tube users 2001-2019')

plt.show()
```



3b) Fit a linear model $f(x)=\beta_0+\beta_1 x$ to the data

- Print the values of the regression coefficients β_0, β_1 determined using least-squares.

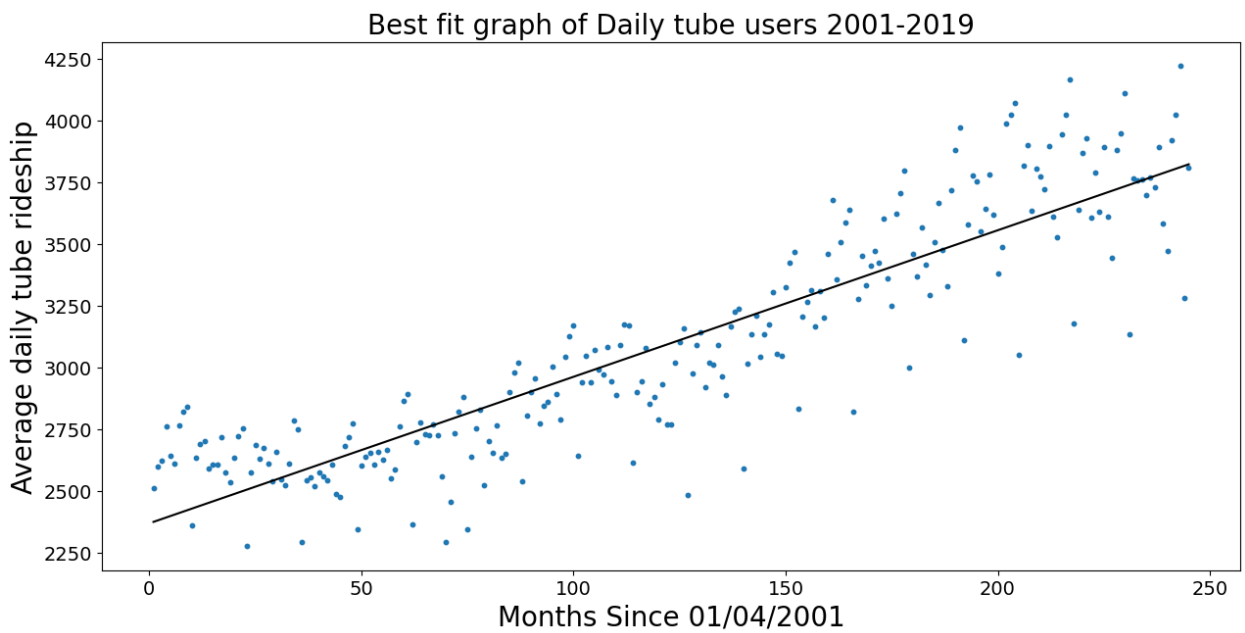
- Plot the fitted model and the scatterplot on the same plot.
- Compute and print the **MSE** and the R^2 coefficient for the fitted model.

All numerical outputs should be displayed to three decimal places.

```
#From Exercise 1
def polyreg(data_matrix, k):

    N = np.shape(data_matrix)[0]

    if k<N:
        order=k
    else:
        order=N-1

    x_data = data_matrix[:,0]
    y_data = data_matrix[:,1]


    X = np.column_stack([x_data**i for i in range (0,order+1)])
    XT= X.T

    betas = np.linalg.inv(XT.dot(X)).dot(XT.dot(y_data))
    fit = X.dot(betas)
    resid = y_data - fit


    #appending additional zeros
    betas=np.append(betas,[0 for i in range(order,k)])
    # The function should return the the coefficient vector beta, the
fit, and the vector of residuals
    return(betas,fit,resid)


data_matrix = np.column_stack([xvals,yvals])
betas,fit,resid=polyreg(data_matrix,1)

print('beta_0 = ', np.round(betas[0], 3), ' \t beta_1 = ',
np.round(betas[1], 3))


plt.scatter(xvals, yvals, s=10)   # s can be used to adjust the size
of the dots
plt.plot(xvals, fit, color = 'black')   # s can be used to adjust the
size of the dots
plt.xlabel('Months Since 01/04/2001')
plt.ylabel('Average daily tube rideship')
plt.title('Best fit graph of Daily tube users 2001-2019')

plt.show()
```

```
fit_0 = np.full_like(yvals, np.mean(yvals))
SSE_0 = np.linalg.norm(yvals - fit_0)**2
SSE_1 = np.linalg.norm(resid)**2
MSE_1 = SSE_1/np.size(yvals)
R2 = 1 - SSE_1/SSE_0

print('SSE_0 = ', np.round(SSE_0, 3), ' \t SSE_1 = ', np.round(SSE_1,
3))
print('MSE_1 = ', np.round(MSE_1, 3), ' \t R^2 = ', np.round(R2, 3))
```

```
beta_0 =  2367.382     beta_1 =  5.939
```

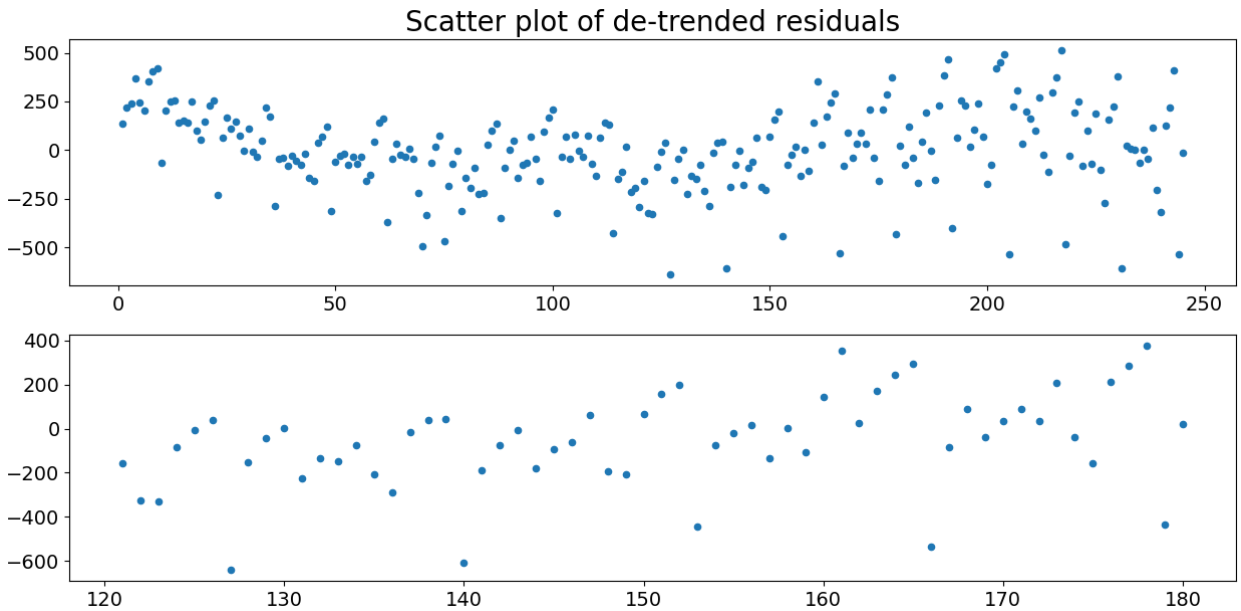



```
SSE_0 =  54329477.0    SSE_1 =  11104290.801
MSE_1 =  45323.636     R^2 =  0.796
```

3c) Plotting the residuals

- Plot the residuals on a scatterplot
- Also plot the residuals over a short duration and comment on whether you can discern any periodic components.

```
plt.subplot(211)
plt.scatter(xvals, resid, s=20)

plt.title('Scatter plot of de-trended residuals')
# Zoom in to a few values
plt.subplot(212)
plt.scatter(xvals[120:180], resid[120:180], s=20)

plt.show()
```

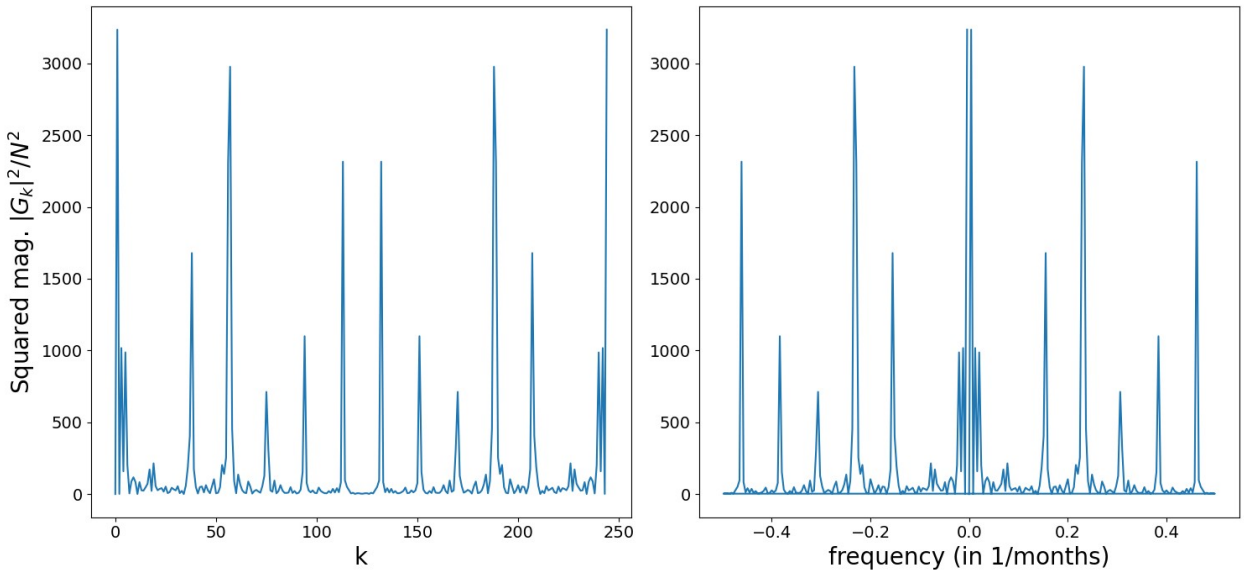


There appears to be a consistent drop (single data point) every 12 month period or so, and the overall shape seems to repeat every year. This is an expected pattern.

3d) Periodogram

- Compute and plot the peridogram of the residuals. (Recall that the periodogram is the squared-magnitude of the DFT coefficients.)
- Identify the indices/frequencies for which the periogram value exceeds **50%** of the maximum.

```
N = np.size(xvals)
T = 1  # period between each data point is 1 month

# Note that T is in months

# Compute the squared magnitudes of the DFT coefficients -- this is
known as the "periodogram"
pgram = np.abs(np.fft.fft(resid, N)/N)**2 #We normalize by N, but this
is optional
indices = np.linspace(0, (N-1), num = N)
freqs_in_hz = np.fft.fftfreq(N)/T
freqs_in_rads = freqs_in_hz*2*math.pi

plt.subplot(121)
plt.plot(indices, pgram)
plt.xlabel('k')
plt.ylabel('Squared mag. $|G_k|^2/N^2$')
plt.subplot(122)
plt.plot(freqs_in_hz, pgram)
plt.xlabel('frequency (in 1/months)')  # Since units of T is months
plt.savefig('DFT_CO2.pdf', bbox_inches = 'tight')
plt.tight_layout()
```




```
top_inds = indices[(pgram > 0.5*np.max(pgram))]
top_freqs_hz = freqs_in_hz[(pgram > 0.5*np.max(pgram))]
print('Top indices:', top_inds, ' Top frequencies in Hz:',
top_freqs_hz)
```

```
Top indices: [  1.  38.  56.  57. 113. 132. 188. 189. 207. 244.]  Top
frequencies in Hz: [ 0.00408163  0.15510204  0.22857143  0.23265306
0.46122449 -0.46122449
 -0.23265306 -0.22857143 -0.15510204 -0.00408163]
```

3e) To the residuals, fit a model of the form

$$\beta_{1s}\sin(\omega_1 x)+\beta_{1c}\cos(\omega_1 x)+\beta_{2s}\sin(\omega_2 x)+\beta_{2c}\cos(\omega_2 x)+\ldots+\beta_{Ks}\sin(\omega_K x)+\beta_{Kc}\cos(\omega_K x).$$

The frequencies $\omega_1,\ldots,\omega_K$ in the model are those corresponding to the indices identified in Part 2c. (Hint: Each of the sines and cosines will correspond to one column in your X-matrix.)

- Print the values of the regression coefficients obtained using least-squares.

All numerical outputs should be displayed to three decimal places.

```
def sinusoid_fit(resid_matrix,freqs_in_hz):

    freqs_in_rads = 2*np.pi*freqs_in_hz
    x_data = resid_matrix[:,0]
    resid_data = resid_matrix[:,1]

    XT = np.vstack([np.vstack([np.sin(w*x_data), np.cos(w*x_data)])
```

```
for w in freqs_in_rads])
    X = np.transpose(XT)

    beta_sc = np.linalg.inv(XT.dot(X)).dot(XT).dot(resid_data) #
Calculating the least-squares coefficients
    #print('The coefficients beta_s, beta_c are:', beta_sc)
    fit_sc = X.dot(beta_sc)
    resid_sc = resid_data - fit_sc

    return(beta_sc,fit_sc,resid_sc)


resid_matrix = np.column_stack([xvals,resid])
freqs_in_hz = np.array([f for f in top_freqs_hz if f>0])
beta_sc,fit_sc,resid_sc = sinusoid_fit(resid_matrix,freqs_in_hz)

for i in range(0,len(beta_sc),2):
    print('freq_'+ str((i//2)+1) +' = ', np.round(freqs_in_hz[i//2],3)
,' 1/month \t beta_'+ str((i//2)+1) +'s = ', np.round(beta_sc[i], 3),
          ' \t beta_'+ str((i//2)+1) +'c = ', np.round(beta_sc[i+1],
3))
```

```
freq_1 =  0.004  1/month    beta_1s =  -51.253    beta_1c =  101.556
freq_2 =  0.155  1/month    beta_2s =  61.628     beta_2c =  -54.006
freq_3 =  0.229  1/month    beta_3s =  -15.581    beta_3c =  -94.797
freq_4 =  0.233  1/month    beta_4s =  81.659     beta_4c =  72.381
freq_5 =  0.461  1/month    beta_5s =  32.472     beta_5c =  90.589
```

3f) The combined fit

- Plot the combined fit together with a scatterplot of the data
- Compute and print the final **MSE** and R^2 coefficient. Comment on the improvement over the linear fit.

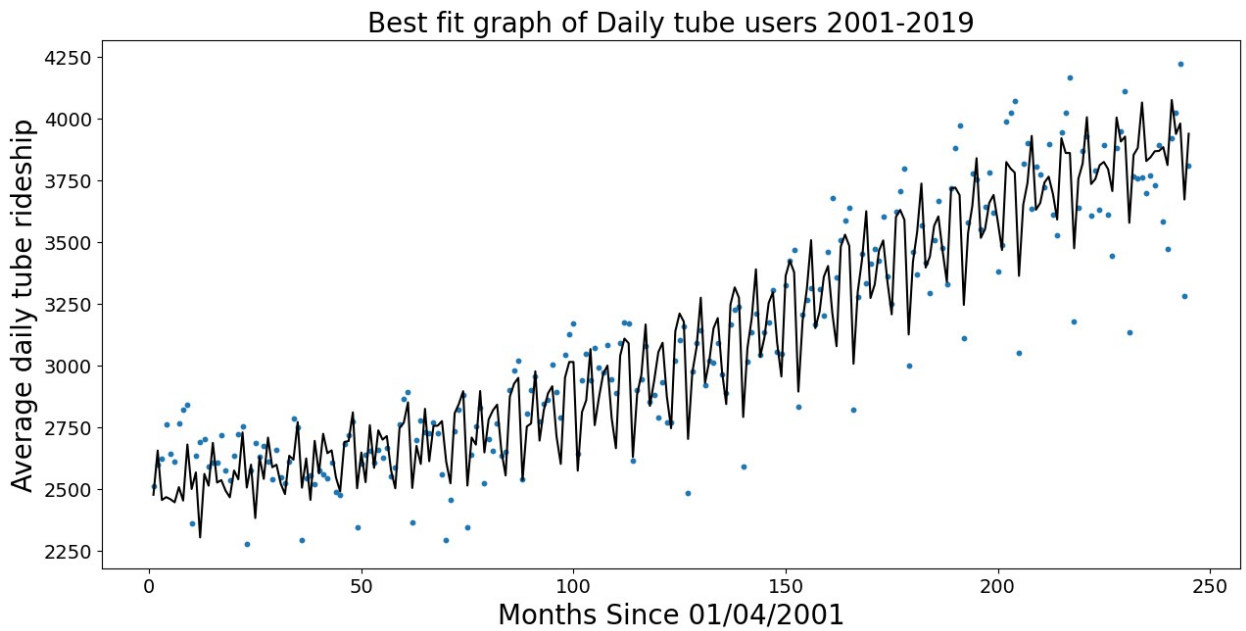
The combined fit, which corresponds to the full model

$$f(x)=\beta_0+\beta_1 x+\beta_{s1}\sin(\omega_1 x)+\beta_{c1}\cos(\omega_1 x)+\ldots+\beta_{sk}\sin(\omega_k x)+\beta_{ck}\cos(\omega_k x),$$

can be obtained by adding the fits in parts 2b) and 2e).

```
total_fit = fit + fit_sc

plt.scatter(xvals, yvals, s=10)   # s can be used to adjust the size
of the dots
plt.plot(xvals, total_fit, color = 'black')   # s can be used to
adjust the size of the dots
plt.xlabel('Months Since 01/04/2001')
plt.ylabel('Average daily tube rideship')
plt.title('Best fit graph of Daily tube users 2001-2019')
```

```
plt.show()

SSE_final = np.linalg.norm(resid_sc)**2
MSE_final = SSE_final/np.size(yvals)
R2_final = 1 - SSE_final/SSE_0

print('SSE_0 = ', np.round(SSE_0, 3), ' \t SSE_sc = ',
np.round(SSE_final, 3))
print('MSE_sc = ', np.round(MSE_1, 3), ' \t R^2 = ',
np.round(R2_final, 3))
```




```
SSE_0 =  54329477.0    SSE_sc =  4972887.791
MSE_sc =  45323.636    R^2 =  0.908
```

A significant improvment in goodness of fit: from R^2 = 0.796 in the linear approximation, to R^2 = 0.908 with in sinusoidal components. The overall shape of the plotted curve also fits the data a lot better visually than the initial linear approximation (which only had a general upwards trend prediction). The sinusoidal inclusion alows the model to predict seasonal variation. Further testing, with greater order polynomial trend and more sinusoidal components, displays a better goodness of fit - however, this increases the complexity of the model.